

Conditional Probability (7.7)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\frac{3}{20}$

_____ Find $P(A \mid B)$ if $P(A \cap B) = 0.1$ and $P(B) = 0.5$.

ANSWER: $\frac{1}{2}$

_____ On a tree diagram, in the bedroom she studies math 70% of the time. Find $P(\text{math} \mid \text{bedroom})$.

ANSWER: $\frac{9}{19}$

_____ Find $P(A \mid B)$ if $P(A \cap B) = 0.1$ and $P(B) = 0.4$.

ANSWER: $\frac{2}{5}$

_____ Find $P(A \mid B)$ if $P(A \cap B) = 0.15$ and $P(B) = 0.5$.

ANSWER: 0.2 # _____ From a table, $P(\text{bus} \mid \text{9th grade}) = \frac{36}{76}$. Simplify.	ANSWER: $\frac{31}{53}$ # _____ Find $P(B \mid \text{red})$ if $P(B \cap \text{red}) = \frac{4}{15}$ and $P(\text{red}) = \frac{2}{3}$.
ANSWER: $\frac{3}{8}$ # _____ Find $P(A \mid B)$ if $P(A \cap B) = 0.3$ and $P(B) = 0.6$.	ANSWER: 0.3 # _____ From a table, $P(\text{12th grade and car}) = \frac{45}{300}$. Simplify.
ANSWER: $\frac{1}{4}$ # _____ In a class, 16 use TikTok and 6 use both TikTok and Instagram. Find $P(\text{Instagram} \mid \text{TikTok})$.	ANSWER: 0.7 # _____ From a table, $P(\text{adult} \mid \text{disliked}) = \frac{31}{53}$. (Already in lowest terms.)